

From V. B. Aleskovskii's "Framework" Hypothesis to the Method of Molecular Layering/Atomic Layer Deposition**

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This essay is dedicated to the history of creation and development of the molecular layering technique (ML) which, in the modern community of non-Russian scientists, is commonly referred to as atomic layer deposition (ALD). Basic research in the field of chemical transformations of solid surfaces using the ML method in the light of the "framework" hypothesis proposed by V. B. Aleskovskii in 1952 is discussed. A number of questions raised by international scientists including those involved in the Virtual Project on the History of ALD (VPHA, 2013), and scientists from conferences in Helsinki (Finland, May 2014.), Kyoto (Japan, June 2014), and personal communications amongst peers are addressed. For the first time in English, this article provides information about V. B. Aleskovskii and S. I. Kol'tsov who are closely associated with development of the ML technique in the Soviet Union. This paper also informs the scientific community about research groups currently engaged in ML research in Russia and introduces the scientific school of "Chemistry of highly organized substances", founded and supervised by V. B. Aleskovskii.

Keywords: ALD, molecular layering, history, "framework" hypothesis

1. Introduction

There are three main names for what is considered to be the same technology of gas-phase layer by layer deposition. Russians commonly refer to the technology as molecular layering (ML), Europeans have traditionally referred to the technology as atomic layer epitaxy (ALE), and most non-Russians now refer to it as atomic layer deposition (ALD). The first publications mentioning the molecular layering technique (ML) appeared in the 1960s.^[1–7] Many of the first publications appeared only in local editions of abstract books from scientific and technical conferences at the Leningrad Technological Institute by Lensovet (LTI by

Lensovet) and were, therefore, difficult to access. For this reason, only a few such references are given in this paper.^[1,8,9] The first full article on the ML method appeared in 1965.^[2] Since the late 1970s, interest in the method of atomic layer deposition (ALD, known in Russia as ML) grew exponentially. This can be seen by the number of publications mentioning ML or ALD between 1972 to 2010 as shown in the reviews^[10,11] by Riikka Puurunen et al.

In recent years there have been several review articles and an essay published about the history of ALD.^[10,12,13] In some of these articles, molecular layering (ML) was briefly mentioned alongside atomic layer epitaxy (ALE) and atomic layer deposition (ALD). For example, in the article, "History of atomic layer deposition and its relationship with the American Vacuum Society" ^[12] the authors write: "The principle of ALD was first published under the name 'Molecular Layering' (ML) in the early 1960s by Prof. S. I. Kol'tsov from the Leningrad (Lensovet) Technological Institute (LTI). . . . the concept of the ALD process was first proposed by Prof. V. B. Aleskovskii in his Ph.D. thesis published in 1952". However, these articles provide no information on the evolution of this research between 1952 and 1977 when the first patent on atomic layer epitaxy (priority November 29, 1974) by T. Suntola (Finland) was published.^[14] Therefore, one of the objectives of this review is to fill this historical gap.

Today the ML/ALD method is one of the fastest growing processes in the world with respect to "core-shell" type material production systems. It is a classic example of a "bottom-up" gas-phase nanofabrication process. It is therefore not surprising that the number of world scientists

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involved in the study and commercialization of the ML/ALD method is also growing exponentially. As more people get involved with the ML/ALD method, more and more questions arise about the history of whom and when it was created, what was the impetus for ML/ALD development, and what the current status and prospects are for the method. Partial answers to these questions have been given in both early and recent publications as well as in the international Virtual Project on the History of ALD (VPHA).^[15–18] The VPHA was initiated by colleagues from Finland (Dr. Riikka Puurunen, Senior Scientist, VTT Technical Research Centre of Finland), Germany (Dr. Jonas Sundqvist, Group Leader High-k Devices, Fraunhofer IPMS-CNT, Germany) and the USA (Dr. Aziz Abdulagatov, Postdoctoral Fellow, National Institute of Standards and Technology [NIST, Boulder], USA).^[18]

Within the VPHA,^[18] the Russian collaborators presented numerous published materials (up to 1974) including dissertations, articles, conference proceedings, monographs, and textbooks on the subject of ML. Russian scientists and their colleagues from other countries were given a chance to comment on all of the publications. Most of the very early publications were previously unknown since they were not readily available to the world scientific community. This lack of access arose from: the political climate of the era, the close-knit ML research community itself, the limited number of articles published by Soviet researchers in international journals, and the fact that in the Soviet Union ML method was originated by chemists, as opposed to physicists in Finland. Soviet information on ML was mainly published in chemistry journals, which was often not of interest to the physicists. Following the VPHA discussions of the project materials, abstracts and conclusions were drawn on the history of ML/ALD by more than 30 participating scientists from 10 countries. These agreed-upon facts were then prepared and presented by R. Puurunen at the ALD conferences in Helsinki (May 2014, Finland) and in Kyoto (June 2014, Japan).^[19,20] Later R. Puurunen published essay on the history of the ALD in Finland.^[21]

Despite the relatively large amount of information that has been presented at such conferences, many colleagues have questions about: the origins of ML in Russia; scientists and research groups engaged in ML research, their research interests and international collaborations; and the use of this technology in industry.

In the context of the origins of ALD in Finland, R. Puurunen noted^[21] that the inventor of the method is T. Suntola (physicist by education). In his first patent of 1977^[14] (priority November 29, 1974, publication November 15, 1977, publication in the USSR February 28, 1981) T. Suntola called his new thin-film deposition method “atomic layer epitaxy (ALE)”. This method was not an analogy with the presently well-known thin film deposition technique “molecular beam epitaxy”.^[22] Since then, Finish and international chemists have joined this research

area.^[23–29] The impulse for the spread of ALE and the beginning of naming it ALD was the making of advanced electroluminescent panels.

Fundamentally, a different technological development path was followed for ML in Russia versus ALD/ALE in Europe. The different path in Russia likely resulted from Russians focusing more on the fundamental chemistry studies and chemical transformations of solids. First Russian ML scientists were chemists from the USSR and graduates of the Leningrad Technological Institute by Lensovet (LTI by Lensovet) Professor V. B. Aleskovskii and his graduate student Professor S. I. Kol'tsov.

This article presents the main directions and stages of formation and development of the ML method in the Soviet Union and, later on, in Russia. Information about the inventors of the ML method is also included. To complement the R. Puurunen essay on “the development of ALD method in Finland by T. Suntola and his colleagues”, we highlight the following sections:

1) 1950s, and early 1960s - the development of the fundamental bases of a new direction in the chemistry of solids; 2) mid-1960s to 1974 - the period before the first Finnish patent (priority 1974); 3) 1975–1980s – the beginning of active development of ML/ALD in the world; and 4) 1990s – present day - the latest advances in ML.

The most detailed analysis of research achievements by Russian scientists is covered in the 1974–1979 period when T. Suntola and his colleagues obtained their patents on the ALE method and when numerous Russian publications on ML appeared. This is the period with which the wider ALD scientific community is least familiar.

2. V. B. Aleskovskii and S. I. Kol'tsov: The Founders of the Molecular Layering Method

In Russia, the inventors of the ML method are Prof. V. B. Aleskovskii (a chemist by education) and his student S. I. Kol'tsov, both graduates of oldest educational chemical engineering institute in Russia, the Leningrad Technological Institute by Lensovet (Fig. 1). This institute was established by the decree of Emperor Nicholas I in 1828 as a “practical institute of technology”. Currently it is called the St. Petersburg State Technological Institute (Technical University). Practically the entire scientific and pedagogical activity of V. B. Aleskovskii and his student S. I. Kol'tsov was related to solid state chemistry. In the walls of this institute between 1950 and the late 1960s they created and experimentally substantiated the principles of the ML method.

Valentin Borisovich Aleskovskii was born on June 3rd 1912 in the city of Mary (Turkmenistan). In 1931, Valentin Borisovich was admitted to the LTI by Lensovet and, after its completion in 1937, was sent to the Naval Research Institute in Leningrad, where he worked as a research fellow. In 1938 he started his graduate studies at the LTI by Lensovet, and in 1940 he defended his candidate of



Fig. 1. Inventors of the molecular layering method V. B. Aleskovskii (left) and S. I. Kol'tsov (early 2000). Photographs from Malygin's collections; published with permission of Anatoly Malygin (private collection).

chemical sciences degree (roughly equivalent to a Ph.D.). His thesis title was "Active silica" and, before the Second World War, he went on to work at the same institution as a senior researcher in the Department of Sorption Technology.

In June 1941 Valentin Borisovich volunteered for the war and, after returning to the institute, he worked as a research assistant in the Department of Physical Chemistry. He received government awards for participation in the Second World War. In 1949 he was appointed as the head of the Department of Analytical Chemistry. In 1952 he defended his doctoral dissertation (Table 1) and became a Doctor of Chemical Sciences and a Professor.¹ In his doctoral dissertation on page 98 V. B. Aleskovskii notes that the original idea of "framework" hypothesis was outlined by him in 1940.

Stanislav Ivanovich Kol'tsov had a similar path to joining the LTI by Lensovet, but he arrived post-war. Stanislav Ivanovich was born on August 30th, 1931 at the station Proletarskaya in the Rostov region of the USSR. In 1950 he was admitted to the Leningrad Technological Institute by Lensovet and he graduated in 1955 with a chemical engineering degree. In the same year, S. I. Kol'tsov enrolled as a graduate student at the Department of Analytical Chemistry. His supervisor was Prof. V. B. Aleskovskii. Since then they fruitfully worked together until the early 21st century.

Originally formulated in V. B. Aleskovskii's doctoral dissertation, the "framework" hypothesis required further

development and further experimental validation. Based on the hypothesis, V. B. Aleskovskii developed a new understanding of solids and their potential for physical and chemical transformations. At that point there was no task to create a thin film synthesis method to solve a particular practical problem, but as part of the basic academic research, a number of various synthesis methods were developed to confirm the underlying principles of the "framework" hypothesis.

The first phase of research under V. B. Aleskovskii's guidance ended in 1963 when S. I. Kol'tsov defended his candidate of chemical sciences thesis (Table 1). The same year S. I. Kol'tsov wrote a monograph based on his work.^[30] From this work there began a systematic study of the interaction of surface functional groups on silica gel and other solid materials (matrices) with chlorides of different elements.

After S. I. Kol'tsov, candidate degree dissertations were defended by G. N. Kuznetsova (1965), A. N. Volkova (1969) and G. V. Sveshnikova (1971) (Table 1).

The final stage of development of the ML method was S. I. Kol'tsov's doctoral dissertation (Defended on January 20, 1971, Table 1), in which he laid out the key principles of the ML method and illustrated its synthetic capabilities and promising practical applications.

Furthermore, before the Finnish patent of 1977 (priority 1974),^[14] work on ML was actively continued as shown in the theses of R. R. Rachkovskii (1972), A. A. Malygin (1973), V. M. Smirnov (1973), T. V. Tuz (1974), and E. P. Smirnov (1974) (Table 1).

By 1979 (second Finnish patent was granted in 1983^[31]), contributions to the development of the molecular layering method were also made by V. B. Aleskovskii's and S. I. Kol'tsov's students, V. B. Kopylov (1977), A. A. Malkov (1977), V. I. Kovalkov (1977), M. A. Ereemeeva (1977), N. A. Stepanova (1978), A. M. Postnova (1978) and V. E. Drozd

¹ In the Soviet Union - Russia, there are two types of scientific degree: 1. Candidate of Sciences (chemical, technical, physico-mathematical); to obtain this scientific degree must submit and defend a candidate dissertation (thesis); 2. Higher scientific degree - Doctor of Sciences (chemical, technical, physico-mathematical); to obtain this scientific degree must submit and defend a doctoral dissertation (thesis).

Table 1. List of selected ML dissertations.

Author and year	Title of the dissertation ^{a)}	Name of the university where the dissertation was made
Candidate dissertations		
S. I. Kol'tsov, 1963	The study of interaction of carbon tetrachloride with silica gel	LTI by Lensovet
G. N. Kuznetsova, 1965	Study of a reaction products between hydrated silica and metal ions by IR spectroscopy	LTI by Lensovet
A. N. Volkova, 1969	Interaction of some chlorides with silica gel - reaction of molecular layering	LTI by Lensovet
G. V. Sveshnikova, 1971	Synthesis and study of thin oxide layers on the silicon surface	LTI by Lensovet
R. R. Rachkovskii, 1972	Synthesis and study of thin oxide layers on the surface of single-crystal silicon	LTI by Lensovet
A. A. Malygin, 1973	Interaction of oxychlorides of vanadium, chromium and phosphorous with silica gel - reaction of molecular layering	LTI by Lensovet
V. M. Smirnov, 1973	Synthesis and activity study of a thin oxide layers on the silica gel surface	LTI by Lensovet
T. V. Tuz, 1974	Synthesis and stability of thin oxide layers on the silica gel surface	LTI by Lensovet
E. P. Smirnov, 1974	Synthesis of carbon and titanium oxide layers on carbon surface by the molecular layering method	LTI by Lensovet
V. I. Kovalkov, 1977	Synthesis of fine powder with desired structural and chemical properties of their surface using molecular layer deposition and the study of their properties	LTI by Lensovet
V. B. Kopylov, 1977	Synthesis of carbon layers on the silica gel surface by molecular layering and study of their properties	LTI by Lensovet
A. A. Malkov, 1977	Synthesis by molecular layering method of metal oxide layers on the surface of the carbon fibers and study of their properties	LTI by Lensovet
M. A. Eremeeva, 1977	Synthesis and research by IR - ATR spectroscopy methods and ellipsometry of ultrathin oxide films on the surface of single crystal silicon and germanium	LTI by Lensovet
A. M. Postnova, 1978	Study of the structure and reactivity of vanadium containing silica gel obtained by molecular layer deposition	LTI by Lensovet
N. A. Stepanova, 1978	Synthesis of iron and zinc-containing layers on the surface of silica gel	LTI by Lensovet
V. E. Drozd, 1978	Synthesis and study of oxide coatings obtained by molecular layer deposition technique on the surface of semiconductors	LTI by Lensovet
V. K. Gromov, 1979	Synthesis by the molecular layering method of titanium oxide layers on the surface of quartz, silicon, copper and ellipsometric study of substrate/ synthesized layer boundary	LTI by Lensovet
V. N. Postnov, 1979	Synthesis of inorganic matrices using the molecular layering method and the study of their reactivity in the process of adsorption of amino acids	LTI by Lensovet
M. N. Tsvetkova, 1980	Molecular layering of active structures on the surface of the glass microspheres and study of their physical and chemical properties	LTI by Lensovet
V. P. Tolstoy, 1980	Synthesis and physicochemical study of ultrathin oxide layers on metal surfaces	LTI by Lensovet
S. K. Gordeev, 1980	Synthesis of carbon, titanium, and chromium oxides on the diamond surface and study of their physico-chemical properties	LTI by Lensovet
B. D. Sokolov, 1980	Effect of physical and chemical properties of carbon fiber surface on the reinforcement of composite materials	LTI by Lensovet
V. D. Ivin, 1980	Synthesis of carbon and of phosphorus-carbon coatings on the carbon fiber surface and the study of their physico-chemical properties	LTI by Lensovet
A. V. Krasnobryzhii, 1981	Reaction of titanium (IV) and iron (III) chlorides, vanadium (V) and chromium (VI) oxychlorides with various graphite materials	LTI by Lensovet
V. F. Dergachev, 1982	Development of technology for gas phase silica gel modification by vanadium	LTI by Lensovet
N. N. Kopylov, 1982	Structure and properties of ultrathin titanium oxide layers synthesized on the monocrystalline silicon surface in an electric field	LTI by Lensovet
A. L. Egorov, 1983	Synthesis of thin oxide films on the surface of tantalum and gallium arsenide, their structure and properties	LTI by Lensovet
A. V. Brykalov, 1983	Synthesis of silica gel based hemosorbents by molecular layering method and their physico-chemical properties	LTI by Lensovet
S. E. Kurashvili, 1984	The reaction of ammonia with a copper-titanium and copper-vanadium films synthesized on the surface of copper	LTI by Lensovet
A. V. Evdokimov, 1985	Synthesis of multicomponent metal oxide monolayers on the silica gel surface and characteristics of their structure and inter-functional interactions	LTI by Lensovet
A. A. Seytmagzimov, 1985	Chemical modification and the surface properties of the anode aluminum and titanium oxides	LTI by Lensovet
T. A. Vitkovskaia, 1985	Surface modification of photoluminophors by the molecular layering technique	LTI by Lensovet
T. B. Lvova, 1986	Chemical modification of coal from the gaseous phase by vanadium (V), chromium (VI), and phosphorus (III) compounds and study of their absorptive properties	LTI by Lensovet
A. B. Zhidkov, 1986	Gas phase kinetics of heterogeneous synthesis of titanium carbide and nitride on the surface of carbon materials	LTI by Lensovet
A. N. Kriulkin, 1987	Physico-chemical properties of multilayer titanium-vanadium oxide films synthesized by molecular layering method on the surface of silicon	LTI by Lensovet

(Continued)

Table 1. (Continued)

Author and year	Title of the dissertation ^{a)}	Name of the university where the dissertation was made
A. S. Yakovlev, 1987	Properties of interface boundaries between monocrystalline silicon and ultra-thin layers of titanium and silicon oxides synthesized by molecular layering method	LTI by Lensovet
V. A. Mitnikov, 1987	Synthesis, formation and some properties of zinc chalcogenide films produced by atomic layering	LTI by Lensovet
A. I. Kudryashova, 1987	Acid-base properties of silicon, aluminum, zinc, and magnesium oxide surfaces and their changes in the process of structural and chemical transformations	LTI by Lensovet
S. A. Trifonov, 1987	Synthesis and thermal-oxidative stability of reaction products of phosphorus trichloride and phenolformaldehyde epoxy materials	LTI by Lensovet
E. Yu. Popova, 1988	Gas phase modification of the dispersed silicon dioxide material by Zn, Fe, Ti containing structures and the use of these synthetic products in polymer composites	LTI by Lensovet
S. V. Murashov, 1988	Synthesis and formation of cadmium chalcogenide films by surface chemical assembly	LTI by Lensovet
V. D. Kupriyanov, 1989	Synthesis on the dispersed silicon dioxide surface and modification of the luminescent zinc sulfide type structures	LTI by Lensovet
E. A. Avrutina, 1989	Physico-chemical properties of the optically active titanium oxide structures synthesized by the molecular layering method on the surface of highly dispersed silica and zinc	LTI by Lensovet
Yu. M. Artemyev, 1989	Physico-chemical properties of iron polysilicates synthesized by chemical assembly	LSU
P. V. Rogowskii, 1990	Physico-chemical properties and reactivity of multilayer vanadium, phosphorus and titanium oxide systems obtained by the molecular layering method	LTI by Lensovet
A. I. Klusevich, 1990	The synthesis and properties of thin film structures based on aluminum and tantalum oxides	LTI by Lensovet
E. A. Sosnov, 1991	Structural-chemical transformations on the surface of silicon dioxide by molecular layering, titanium - and silicon-nitrogen structures in the temperature range 200–800°C	LTI by Lensovet
Zh. N. Ishutina, 1997	Phase formation and properties of materials in the compositions on the basis of the system $\text{Al}_2\text{O}_3\text{-SiO}_2\text{-TiO}_2$	SPbSTI
S. D. Dubrovenskii, 1997	Synthesis of vanadium titanium oxide nanostructures on the surface of silica gel and pyrographite and simulation of the processes of their formation	SPbSTI
O. V. Osipenkova, 1998	Local physico-chemical transformations on the surface of silica in the processes of interaction with TiCl_4 , VOCl_3 , CrO_2Cl_2 и H_2O	SPbSTI
A. I. Kutchiev, 2007	Synthesis and quantum-chemical study vandieken structures on the silica surface and their interaction with pairs of VOCl_3 and H_2O	SPbSTI
A. K. Bulkina, 2009	Structural and chemical transformations on the surface of films of polyethylene and polyvinyl chloride by reaction with halides of phosphorus, vanadium, titanium and silicon	SPbSTI
Y. M. Koshtyal, 2011	Solid-state structural-chemical transformations in the interaction of porous silica with vapors of TiCl_4 и H_2O	SPbSTI
A. N. Gukova, 2013	Synthesis and quantum - chemical analysis of Ti, V, Cr - containing structures on the surface of the silica	SPbSTI
K. S. Anisimov, 2013.	Synthesis of metal-oxide structures on the surface of the silicon carbide with the use of chloride of titanium, oxohalides chromium and vanadium	SPbSTI
Doctoral dissertations		
V. B. Aleskovskii, 1952	Framework hypothesis and experience the preparation of certain active catalysts	LTI by Lensovet
S. I. Kol'tsov, 1971	Synthesis of solids by the Molecular Layering Method	LTI by Lensovet
V. N. Pak, 1982	Element oxide polyhedral structure and physicochemical properties of the surface oxides	LTI by Lensovet
D. Damyanov, 1988	Growth by molecular layering of a catalytically active phase on the oxide surfaces	Burgas Institute of Technology, Bulgaria
A. A. Malygin, 1991	Theoretical and experimental bases of surface modification technology of dispersed and porous materials by the molecular layering method	SPbSTI
G. U. Ostrovidova, 1992	Physicochemical bases of making carbon containing composite materials	SPbSTI
V. M. Smirnov, 1994	Chemistry of surface chemical compounds of polydioxide silica (dispersed silica gel)	SPbSU
A. P. Nechiporenko, 1995	Donor-acceptor properties of the surface of solid oxides and chalcogenides	SPbSTI
V.V. Gusarov, 1996	Statics and dynamics of polycrystalline systems based on refractory oxides	SPbSTI
A. G. Syrkov, 1998	Hydride solid state synthesis of metallic materials and its basic laws	SPbSU
A. P. Alekhin, 1999	Chemical and technological bases of low temperature interphase boundaries formation between insulator and semiconductor	Research Institute of Physical Problems by N.F. Lukin
Y. K. Yezhovskii, 2002	Correlation analysis in surface chemistry of solids and matrix synthesis of nanostructures	SPbSTI
V. B. Kopylov, 2007	Chemical structure and the role of electron-excited states in the properties of oxides	Saint - Petersburg State Pedagogical University
V. P. Tolstoy, 2009	Synthesis of inorganic nanolayers using "layer by layer" approach at the solid-liquid interface	SPbSU
M. M. Sychev, 2013	Scientific basis for controlling the properties of composite films for electroluminescence devices	SPbSTI

^{a)} Information about of the dissertations can be requested from the Russian State Library (Moscow), in the libraries of the organizations in which research was carried out, on the world wide web and by contacting the authors of the dissertations directly.

(1978), V. K. Gromov (1979), V. N. Postnov (1979) (Table 1).

Today the total number of defended candidate and doctoral theses related to the molecular layering method are more than a hundred. In the Soviet Union, later Russia, the leading scientific school called “Chemistry of highly organized substances” was formed, of which V. B. Aleskovskii was the organizer and the permanent head of school until the last days of his life. It should be noted that the creators of the “framework” hypothesis and the molecular layering method paid considerable attention, not only to scientific work, but also to the educational process, and also actively participated in public affairs.

From 1965 to 1975 V. B. Aleskovskii was acting rector of the LTI by Lensovet. Fig. 2 shows a photograph of his rector identification card (ID). In 1967 he founded one of the first in USSR the Department of Chemistry of Solids and became its chair. In 1989 this department changed its name to the Department of Chemical Engineering of Materials and Electronics Products, and since 2008 it has been known as the Department of Chemical Nanotechnology and Electronics Engineering Materials (CNEEM). In 1972 V. B. Aleskovskii was elected a corresponding member of the Academy of Sciences of the USSR. Since 1975, educational, scientific and organizational activities of V. B. Aleskovskii were associated with the Leningrad State University (LSU) since he was appointment as rector of this university. From 1976 to 1982, the Solid State Chemistry Department at LTI by Lensovet was headed by S. I. Kol'tsov. Fig. 3 shows a photograph of the members of the Department of Chemistry of Solids of the LTI by Lensovet and Fig. 4 shows the members of the molecular layering research group in this department.

In 1977 V. B. Aleskovskii organized the Department of Solid State Chemistry at the LSU which he supervised until 1986 and then worked there as a professor. Figures 5 and 6 show V. B. Aleskovskii with departmental staff and students in one of LSU's laboratories. The main area of scientific interest for V. B. Aleskovskii was the study of chemical transformations of solids and the development of technologies related to new inorganic materials. He is the author of over 450 scientific publications including 12 books and monographs and more than 150 inventions. Review of his early works is published in the Journal of Applied Chemistry.^[32]



Fig. 2. V. B. Aleskovskii's LTI by Lensovet rector ID card, V. B. Aleskovskii (1966). Photograph from Malygin's collections; published with permission of Anatoly Malygin (private collection).



Fig. 3. Collective photograph of the members of the Department of Chemistry of Solids of the LTI by Lensovet (S. I. Kol'tsov, Department Head, sits at the center, 1974). Photograph from Malygin's collections; published with permission of Anatoly Malygin (private collection).



Fig. 4. Members of the molecular layering research group at the Department of Chemistry of Solids at the LTI by Lensovet (from left to right: A. A. Malygin, T. V. Tuz, V. M. Smirnov, S. I. Kol'tsov, V. B. Kopylov, E. P. Smirnov; sitting second from left is A. N. Volkova, 1973). Photograph from Malygin's collections; published with permission of Anatoly Malygin (private collection).



Fig. 5. Members of the Department of Solid State Chemistry during its 25th anniversary (October 2003). Sitting from left to right: E. G. Zemtsova, Y. M. Artemyev, V. B. Aleskovskii, I. V. Murin (department chair), V. M. Smirnov, V. N. Postnov; in the last row, second and third from the right, V. E. Drozd and V. P. Tolstoy. Photograph from Smirnov's collections; published with permission of Vladimir Smirnov (private collection).



Fig. 6. Rector of the Leningrad State University V. B. Aleskovskii (on the right) in the laboratory of the Department of Solid State Chemistry (first and second from the left A. I. Romanychev and V. M. Smirnov, 1983). Photograph from Smirnov's collections; published with permission of Vladimir Smirnov (private collection).

Based on his monograph "The stoichiometry and synthesis of solid compounds" (1976),^[33] V. B. Aleskovskii prepared the textbook "Chemistry of solids" (1978).^[34] Lecture notes from his course entitles "Course of chemistry of supramolecular compounds" (1990) were the basis for the monograph "Chemistry of supramolecular compounds" (1996).^[35] The last released textbook was "Chemical-informational synthesis" (1997).^[36] These publications formed a new solid state concept – the chemistry of highly organized substances.^[37,38]

For many years V. B. Aleskovskii was chairman of the Council of Rectors of Leningrad and a member of the Supreme Soviet of the Russian Federation (SFSR) where, at various times, he headed the Commissions on Education, Youth and others. He was one of the leading organizers of the Leningrad Scientific Center of the USSR Academy of Sciences which in currently headed by Nobel laureate and Academician of the Russian Academy of Sciences (RAS) Zhores I. Alferov.

The scientific school of V. B. Aleskovskii regularly carried out international and Russian conferences including both the conference on "Chemistry of highly organized substances and scientific basis of nanotechnology" (1996, 1998, 2001, 2004) at the St. Petersburg State University and the conference "Surface chemistry and nanotechnology" (1999, 2002, 2005, 2009, 2012) at the St. Petersburg State Technological Institute (Technical University).²

Between 1978 and 1982 the research on synthesis of solid materials using techniques developed under V. B. Aleskovskii's supervision were carried out as part of the inter-university scientific and technical program called "Chemical assembly of materials". The origin of this program is interesting: by the mid-1970s significant scientific bases that allowed sufficient grounds to argue about the

prospects of molecular layering method (when applied to create solid-phase materials with different functionalities) were created in the USSR. In November 1970 V. B. Aleskovskii made a report titled "Obtaining materials with desired properties" at the General meeting of the branch of physical chemistry and technology of inorganic materials of the Academy of Sciences of the USSR. At the same time came understanding that one of the most important areas of application of this new precision technology was electronics. Back then, this was seen as a breakthrough for Russian science which could potentially bring the USSR onto the world stage for advanced electronic manufacturing. However, not all government officials and scientists at the Academy of Sciences of the USSR supported and understood V. B. Aleskovskii's work. Moreover, many prominent scientists believed that it was impossible to create structure with atomic/molecular precision, and did not understand the basic principles behind molecular layering.

In 1971, V. B. Aleskovskii and S. I. Kol'tsov attempted to patent the ML process (application № 1642785/23-26, April 07, 1971). The patent application was called "The principles of synthesis of solids". It was declined by the patent office. This remarkable fact shows that authors were not supported by the scientific community and were ahead of their time. Nevertheless, after one of the V. B. Aleskovskii's articles on chemical assembly in the leading newspaper "Pravda" (1977) the Prime Minister of the USSR, N. A. Kosygin became interested in the subject of ML. N. A. Kosygin then ordered establishment of a committee comprising of members from the Academy of Sciences of the USSR, the Russian Ministry of Education and the State Committee on Science and Technology. V. B. Aleskovskii and his students S. I. Kol'tsov, A. A. Malygin and A. M. Shevyakov attended the final committee meeting, at which V. B. Aleskovskii raised the question of establishing an Institute of Chemical Assembly of Materials so the Soviet Union could retain its leadership in the new ML field. However, this idea was rejected by the committee. In V. B. Aleskovskii's concluding remarks, he said that the existing research teams at LTI by Lensovet and LSU would continue ML studies, but that research would not be enough to reach industrial-scale ML production of electronics. He added that, without the proposed Institute of Chemical Assembly of Materials, the Soviet Union would be forced to purchase ML-based technologies from abroad in the near future. This statement caused confusion and further debate amongst the commission members. After that, the Deputy Minister of Education, E. K. Kalinin, proposed a compromise to V. B. Aleskovskii's new proposed institute and instead proposed to create a new inter-university program under V. B. Aleskovskii's supervision. This inter-university program would be led by LSU and LTI by Lensovet as the most prominent organizations in the ML field. So, in 1978, the Ministry of Higher Education of the Russian Soviet Federative Socialist Republic (RSFSR) founded the "Chemical assembly of materials" inter-university program

²Information about of the conferences can be requested from Russian State Library (Moscow), in the libraries of the SPbSTI, SPbSU, where are books with abstracts of reports.

headed by V. B. Aleskovskii, the leading researcher in the chemistry and technology of solid materials by ML and other precision synthesis methods.

As time went on, V. B. Aleskovskii was proven correct as Russia started, and continues to purchase, ML/ALD technologies from foreign manufacturers. Today there is no doubt that non-Russian entities are the current leaders in both the science and industrial application of ML/ALD technologies. It is important to note that the original ML studies had an important place in V. B. Aleskovskii's research activities but his research was mostly associated with the theories behind manipulating the chemistries of solids and surfaces.

Valentin Borisovich Aleskovskii never stopped working, and tirelessly moved toward the development of theoretical and applied chemistry and chemical transformations of solids and materials. In last years of his life, his research was focused on the development of theories relating to chemical-information and quantum synthesis.^[36,37,39] These concepts were described in article, published after his death in 2006, and are dedicated to the theory of quantum chemical technologies that include a practical approach to the synthesis of nano-sized active carbon.^[39,40] Today it is clear that many theories proposed by V. B. Aleskovskii in the field of synthesis of solid state materials have been since commercialized for nanotechnology applications. Most of these technologies focus on the synthesis of materials with pre-defined stoichiometry and functionality and are mostly in the field of nanotechnology. Until the last days of their lives, S. I. Kol'tsov (passed away May 25, 2003) and V. B. Aleskovskii (passed away January 29, 2006) continued to work in what are among the oldest Russian universities, the well-known classical University (LSU - SPbSU) and the practical Institute of Technology (LTI by Lensovet - SPbSTI).

3. 1950s to Early 1960s: The Development of the Fundamental Bases of a New Direction in the Chemistry of Solids

By the late 1950s and early 1960s, chemists of the scientific school of V. B. Aleskovskii were starting to understand the need for development of new solid state synthesis methods in order to experimentally study the "framework" hypothesis theorized by V. B. Aleskovskii in 1952. Often colleagues from other countries would ask questions such as: How and when did the ML method first appear? What is the relationship between the ML method and the "framework" hypothesis? Is there a definition of the ML method? The answers to these questions can be found in the dissertations, scientific articles, and abstracts of conferences dating between 1952, when the "framework" hypothesis was first formulated in V. B. Aleskovskii doctoral dissertation, and 1963, when S. I. Kol'tsov candidate dissertation was published.

As noted by V. B. Aleskovskii in his "Stoichiometry and Synthesis of Solid Compounds" book,^[33] D. I. Mendeleev (one of the greatest Russian scientists and creator of the Periodic Table of Elements) pointed out that the atoms on the surface of a solid are very different from atoms in the bulk, and that this difference will affect the solid's surface properties. In the same publication V. B. Aleskovskii analyses discussions by early 19th century chemists K. L. Berthollet and J. L. Proust. These discussions debate whether one should consider solids as compounds with variable compositions which do not obey the fundamental laws of chemistry (relating to the laws of definite composition and multiple proportions), or whether one could create an environment of strict reproduction of composition of a solid based on the postulates of classical chemistry.

With these discussions and arguments in mind, V. B. Aleskovskii analyzed the work of many other scholars. Although accumulation and analysis of this knowledge of solids and their transformations lead to further understanding, it did not lead specific answers to several of the main fundamental questions. The most important question left unanswered was: Can one create a solid with strict, pre-specified, composition and structure? This became the fundamental question that motivated V. B. Aleskovskii. Based on the analysis of the scientific literature covering a long period, the first step was to suggest basic approaches to building solids with predetermined composition, which would lead to the expression of pre-defined properties.

Consequently, in his doctoral dissertation of 1952,^[33,41] V. B. Aleskovskii proposed a new solid state concept. According to the "framework" hypothesis, the chemical structure of any solid (amorphous or crystalline, organic or inorganic) can contain a distinct "core" – a superpolyatomic radical consisting of a structural units III, integrally connected by covalent (ionic, metal) bonds and surrounded by functional groups A1, A2, A3, etc. (Fig. 7). The formula of a solid as proposed by V. B. Aleskovskii's chemical model can be represented as:

$$[\text{W}]_a\text{A} \quad (1)$$

In formula (1), the stoichiometric coefficient "a" indicates the amount of chemical equivalents of "core" per one chemical equivalent of surface functional groups. He suggested classification of solids by their core dimensions that can be zero-, one-, two- or three-dimensional.^[33,41] With this approach to understanding solids, chemical transformations can occur in two main ways: by either transforming the core, or by transforming the surface functional groups. Reactivity of the core is substantially less when compared to the reactivity of surface functional groups.

This reactivity difference can be clearly seen in case of heterogeneous catalysts, sorbents. It is also important to note that, according to V. B. Aleskovskii, a solid from a chemist's stand point may be considered as a gigantic

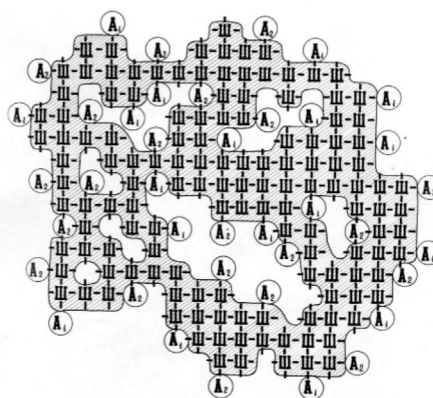
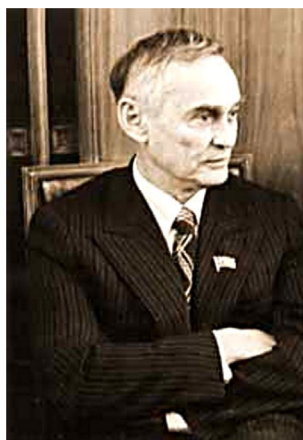


Fig. 7. The chemical model of a solid as proposed by V. B. Aleskovskii's "framework" hypothesis. Photograph from Malygin's collections; published with permission of Anatoly Malygin (private collection). Reproduced with permission.^[41] Copyright 1952, LTI by Lensovet.

molecule. V. B. Aleskovskii introduced the term "supramolecule" for solids. Viewing the solid as a single gigantic molecule allows chemical reactions involving the solid to obey the laws of classical chemistry.^[33,41] Thus, the "core" of a solid body can be represented as macroradical, where the bonds on the surface are occupied by functional groups (atoms or molecules).

A classification system was then developed for a series of solid compounds. These solid compounds had either different core molecular weights and formed a homologous series of solid compounds, or the solids had a constant core molecular weight, with various surface functional groups (pseudomorphs) and formed a genetic series of solid compounds.^[33,41] Each member of such a homologous or genetic series, was viewed as a new substance that had different compositional, structural and physical properties. This means that even addition of a single, chemically-bonded atom to the surface of a solid matrix designated that whole solid a new substance with new properties. The main issue was measuring such minute property changes. At the time, the research tools and methods available to study composition, structure and properties were not available. Since then, the development of nanotechnology has afforded a substantial advance towards measure such minute changes in properties.

Apparently, V. B. Aleskovskii, was the first person to ask whether one could synthesize a solid state structure as an individual chemical compound with constant composition. To test this, new methods were required to prepare surfaces of the original matrix (monocrystalline or amorphous), i.e. position functional groups with given chemical composition and structure on a surface in a certain order; secondly - develop methods to synthesize a solid in a several separate irreversible stages. V. B. Aleskovskii named this new area "supramolecular chemistry of compounds".^[33,35] According to V. B. Aleskovskii, supramolecular chemistry studies the nature of solid compounds where solids are viewed as highly organized supramolecular superpolyatomic individual substances, the synthesis of which is associated with the

making of large amount of chemical information and can therefore only take place in a multi-programmable irreversible process.^[33,35,42,43]

The "framework" hypothesis suggests two main options for chemical transformations of solids. The first option is to modify the functional groups on the surface of a solid matrix. "Core" atoms would not be affected in this option. The second option is transformation of the "core" of the solid as a result of substitution of bulk atoms. Thus, the research was split into two areas, one working on modifying the surface functional groups and one working on modifying core atoms. Reactions of functional groups were studied by S. I. Kol'tsov and, by the mid 1960s, this had become known as "molecular layering" (ML). These surface modifications were carried out predominantly by the supply of gas-phase reactants to the solid's surface. The gas-phase reactants then reacted with, and modified, the solid's surface functional groups.^[1-7,33,35,43] The second option of modifying the core atoms was developed by another of V. B. Aleskovskii's graduate students, Augusta Petrovna Dushina, and this method was called the "method of destructive-epitaxial transformations" (DET).^[44] This method would typically proceed by adding the solid to a liquid reactant.

In his candidate dissertation, S. I. Kol'tsov further studied the "framework" hypothesis by experimenting with carbon tetrachloride interactions on silica gel. It is in this thesis, and related monograph^[30] that molecular layering method was reported at atmospheric pressure. The reactions studied focused mostly on using chloride and oxychlorides various elements reacting with surfaces of silica gels, monocrystalline silicon, quartz, and other types of carbon substrates and matrices. The picture above, taken from page 44 of S. I. Kol'tsov's dissertation, shows a diagram of the experimental setup used for these studies (Fig. 8).

Although there is no mention of the term "molecular layering" in S. I. Kol'tsov's thesis and related book,^[30] the authors suggest the possibility of further transformation of the modified silica after reacting it with certain reagents. In particular, when presenting the data on silica gel

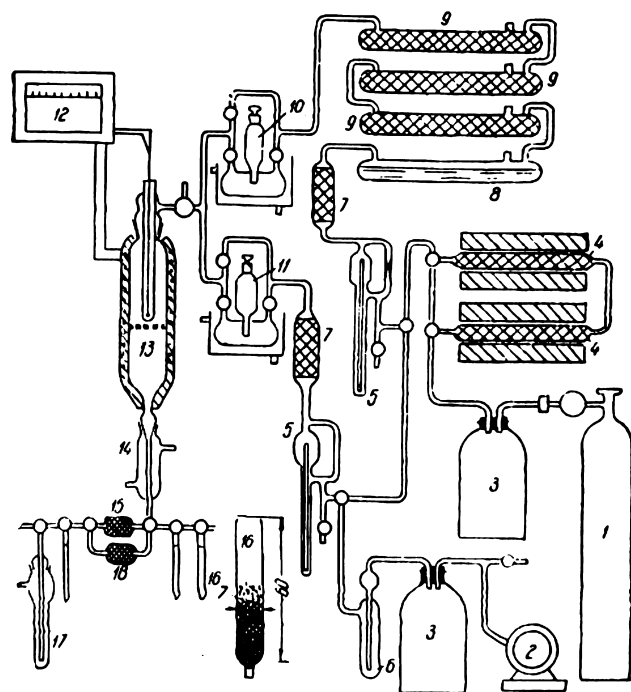


Fig. 8. Schematic of the first molecular layering, flow-type experimental setup by S. I. Kol'tsov. Where 1 is the nitrogen tank; 2 is the blower; 3 is the buffer vessel; 4 is the oven with absorber for separating nitrogen from oxygen impurities; 5 is the rheometer; 6 is the CO_2 absorber; 7, 8, 9 are H_2O absorbers (silica gel, concentrated sulfuric acid, phosphorus pentoxide); 10, 11 - are boats with CCl_4 and H_2O ; 12 is the regulating potentiometer; 13 is the quartz reactor; 14 is the refrigerator; and 15 to 18 are absorbers (granulated zinc powder, alkali silica gel, solution of barium hydroxide, granulated antimony sulfide). Reproduced with permission.^[43] Copyright 1971, LTI by Lensovet.

modification by reaction of its surface OH groups with different precursors, they noted that this way of producing silica "...with a new set of functional groups and thus with new properties" "...contains great potential because it enables reactions with a new functional groups to be carried out and affords a variety of reactions with organic and organoelement compounds which have a great practical importance in the producing new rubber fillers, selective sorbents and catalysts".^[30, p.19–20] Later, in the same publication on page 27, when presenting data on the reaction of trimethylchlorosilane with silica gel (which results in formation of the surface silicon-methyl functional groups) the authors note that "...by removing organic ligands from the silicon-methylated surface of silica gel, and by supplying water to the liberated silicon atoms one arrives at a regenerated hydrated silica matrix...". Therefore, using the "framework" hypothesis as a model for solids, by reacting its functional groups to form new active centers on the surface, a new supramolecular compound is synthesized. This surface may interact with other reactants the newly formed active centers. Furthermore, the process generates a new solid compound at each stage, where each new solid compound has a predetermined set of functional groups.

It is interesting to note that on page 140 of G. N. Kuznetsova's candidate dissertation (1965, Table 1)

(advisors: V. B. Aleskovskii and A. M. Shevyakov - a specialist in infrared spectroscopy) she writes: "[for] studying the reactions of silica gel with HSiCl_3 , TiCl_4 and GeCl_4 we adopted S. I. Kol'tsov's method [that he] developed and used to study interactions with carbon tetrachloride" and she references his thesis work.

We believe that the history presented in this section's discussion above will help answer the question many scientists have about the relationship between the "framework" hypothesis and the method of molecular layering (ML). In the period under review (1952–1963), there were no scientific papers on molecular layering in leading USSR journals and publications are mostly presented in the abstract books of scientific and technical conferences of LTI by Lensovet. In this time period, V. B. Aleskovskii and S. I. Kol'tsov created a step by step fundamental base for further development of solid state chemical modification involving surface functional groups. V. B. Aleskovskii's doctoral thesis, S. I. Kol'tsov's candidate thesis (under the guidance of V. B. Aleskovskii), and their joint monograph^[30] served as the basis for subsequent work on molecular layering performed by later scientists and students.

4. Mid-1960s to mid-1970s (including 1974): Establishment of the Molecular Layering Method and Experimental Studies of its Synthetic Capabilities

Between 1963 and 1974, V. B. Aleskovskii's and S. I. Kol'tsov's students carried out research and development specifically related to the ML method. They studied its features, synthetic capabilities, and how to identify synthesized products. The candidate theses of both G. N. Kuznetsova (1965) and A. N. Volkova (1969) (Table 1) should be noted as being of particular importance. These two theses are the first to lay out principles of synthesis of ML layers of phosphorus, silicon (Fig. 9), titanium, and germanium oxides on silica gel surfaces. They also deposited titanium-phosphorus oxide films using the appropriate chlorides and water vapor as reactants. They also demonstrated the effectiveness IR spectroscopy for the evaluation of surface chemical bond formations in these surface structures.^[45–47] Interestingly, parallel to the ML method work on inorganic materials, similar work emerged on the solid-phase synthesis of polypeptides. This work focused on the surface of polymer substrates under the guidance of R. B. Merrifield.^[48] For his peptide work, R. B. Merrifield was awarded the 1984 Nobel Prize in chemistry.

S. I. Kol'tsov's doctoral dissertation (1971, Table 1) was the final step in establishing the molecular layering method as a sound surface modification technique. In this dissertation, he defined what the molecular layering method was and formulated its principles (p. 219–221). He showed its synthetic capabilities (p. 221–227, schemes 3.1, 3.2, etc.) and proposed a classification system for ML reactions (p. 226–

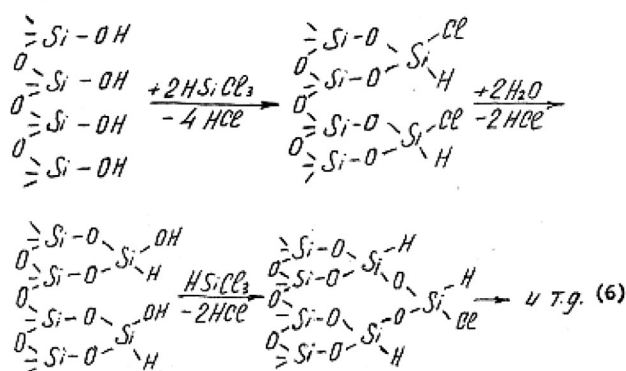


Fig. 9. Schematic representation of the surface reaction of HSiCl_3 and H_2O on silica gel surfaces during the ML process. Reproduced with permission from candidate dissertation of A. N. Volkova. Copyright 1969, LTI by Lensovet.

229). The dissertation also introduced the term “chemical assembly” by the molecular layering method. Based on analysis of the data presented in S. I. Kol'tsov's doctoral thesis, the following should be noted as of importance: The basic idea of the ML method is to sequentially build monolayers of structural units with a given chemical composition and structure on the surface of a solid substrate (matrix) to precisely synthesize solids with regular structure.

To carry out this process there are number of principles that need to be considered: 1) The ML synthesis is based on irreversible chemical reactions between functional groups on the surface of a solid and the molecules of the supplied gas-phase reactant. The reagents and reaction products should not chemically react with each other. 2) For the gradual build-up of the new layer on the solid's surface, repeated and alternating exposures of the reactants must be performed. Each newly formed monolayer should contain the active atoms (or groups of atoms) capable of reacting with the following reagent. 3) For carrying out of reactions by ML there must be some structural correspondence between the surface of starting solid matrix and a given compound. A sufficient number of functional groups with an arrangement that allows cross-linking between atoms is vital to creating a three-dimensional lattice of the synthesized solid.

In S. I. Kol'tsov's doctoral dissertation (p. 229), he classified the reactions of molecular layering into a) uniform; b) uniformly alternating; c) inhibited. Based on S. I. Kol'tsov's thesis data and related work, Fig. 10 shows a diagram of the processes of surface chemical assembly by ML. These ML processes can be used to synthesize nanostructures with different chemical compositions including monolayers and multicomponent layered architectures on solid substrates. Atomic level chemical assembly of surface structures by alternative chemical reactions with a given program (order) can be achieved.

One of the main requirements for reproducible ML synthesis is that the various reactions of the gas-phase precursors (AC_4 , AB_4 , DB_4 , MC_4) with the solid's surface functional groups (B, C) must be carried out far from

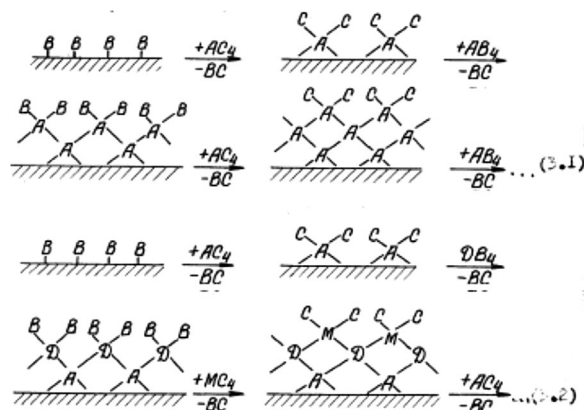


Fig. 10. Synthesis of solids with simple (3.1.) and complex (3.2) chemical composition and structure by the method of molecular layering. Reproduced with permission.^[43] Copyright 1971, LTI by Lensovet.

thermochemical equilibrium. This principle is fundamental because failure to uphold this requirement can lead to uncontrolled synthesis. Possible reaction mechanisms for one ML cycle of chloride and water reactions on a silica gel substrate is presented in Fig. 11. As suggested by this figure, alternative reaction pathways (Routes 3-6, 8) could occur that deviate from the main reaction pathway (Routes 1,2). Routes 3-6 and 8, for example, are associated with the exposure (to the surface) of hydrogen chloride, a byproduct of the either the main reaction between surface hydroxyls on the silica and the gas-phase chloride reactant or from direct chlorination of the matrix surface (reaction 7). Reaction paths 9 and 10 arise from the possible hydrolytic instability of the Si-O-E linkages.

Therefore, if the key ML principles are ignored such as the necessity to be far from thermochemical equilibrium and specific temperature and gas concentration requirements, various side reactions may occur during the attempted ML process. These side reactions may lead to irreproducibility of desired composition and final film structure.

Research into the synthesis of solids by ML significantly intensified after S. I. Kol'tsov's doctoral dissertation. Between 1971 and 1974 six candidate degree theses were defended where S. I. Kol'tsov and V. B. Aleskovskii had joint mentorship: G. V. Sveshnikova (1971), R. R. Rachkovskii (1972), A. A. Malygin (1973), V. M. Smirnov (1973), T. V. Tuz (1974) E. P. Smirnov (1974) (Table 1). The results of this research was in the respective authors' first publications.^[50-58] The works of G. N. Kuznetsova, A. A. Malygin, V. M. Smirnov et al.^[4,59,60] employed infrared spectroscopy and basic analytical chemistry techniques to investigate surface reaction mechanisms and the compositions of ML films. Ellipsometry was used to determine the thickness of these ultrathin coatings produced on both single-crystal silicon wafers (Fig. 12) and quartz surfaces (works by R. R. Rachkovskii, G. V. Sveshnikova, etc.).^[50-52,61]

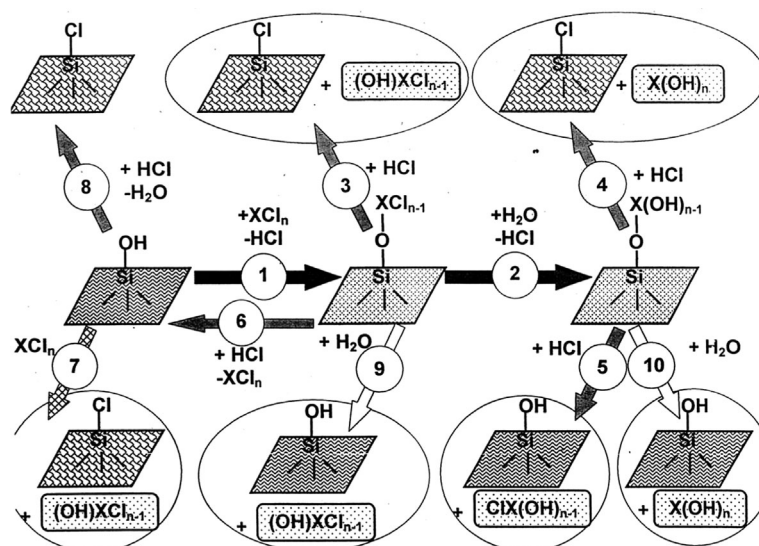


Fig. 11. Possible reaction routes of a silica surface with a chloride precursors. 1,2 - one ML cycle; 3-8 surface chlorination routes; 9,10-hydrolytic degradation pathways.

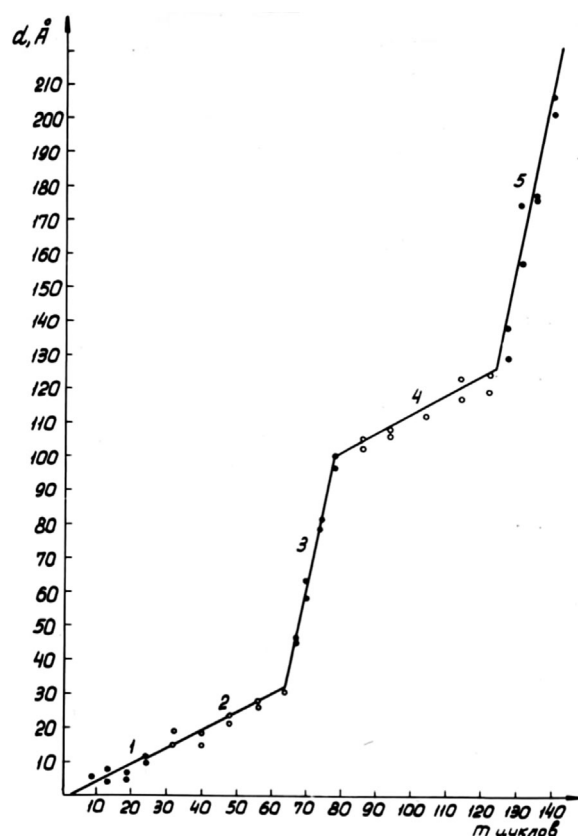


Fig. 12. Ellipsometry film thickness vs. cycle number. Growth of alternating films of TiO_2 and SiO_2 (nanolaminates) on the surface of single crystal silica (five zone system): 24 cycles of TiO_2 (1) - 40 cycles of SiO_2 (2) - 14 cycles of TiO_2 (3) - 44 cycles of SiO_2 (4) - 18 cycles of TiO_2 (5). Reproduced with permission from candidate dissertation G. V. Sveshnikova. Copyright 1971, LTI by Lensovet.

It is important to note that the work carried out between 1963 and 1974 experimentally demonstrated practically all the synthetic possibilities of the ML method as proposed in Fig. 10. Substrates used in this published work include: porous silica gel with high surface area, semiconductor-grade silicon wafers, Ge, quartz, and carbon (activated carbon). Precursors used were predominantly high vapor pressure liquid chlorides and oxychlorides of titanium, vanadium, phosphorus, chromium, boron (boron bromide), silicon (various silanes), germanium, tin, and carbon.^[2-5,7,45,46,50,52-54,57,59-62,90] Depending on the desired chemical composition of the synthesized ML layers the second precursors used were either water (for oxide layers), ammonia (for nitrides), or methane (to obtain carbon films when reacted in pair with carbon tetrachloride).^[4,52,63-65] Aside from condensation reactions of halides with hydroxyl groups, there were groundbreaking ML studies that employed surface redox reactions (Fig. 13) synthesize multicomponent oxide alloys and multilayer films containing elements such as P-Ti (Fig. 14), P-V, Ti-Cr, Cr-V, Cr-P etc.^[66-72]

Between 1965 and 1974, not only fundamental data on the stoichiometry of the synthesized films, but also the results of functional properties of the resulting “core – surface structure” type composites were published.^[72-77] In the period under review, all experimental studies were carried out using atmospheric pressure flow-type ML reactor designs originally proposed by S. I. Kol'tsov (Fig.9). In January 1974, the ML research field was joined by V. E. Drozd, a graduate of the Leningrad Electro-technical Institute (LEI) and a graduate student of the Department of Chemistry of Solids in LTI by Lensovet. He was one of the few physicists in the ML group and was the first person to make a manually controlled vacuum ML reactor.^[78]

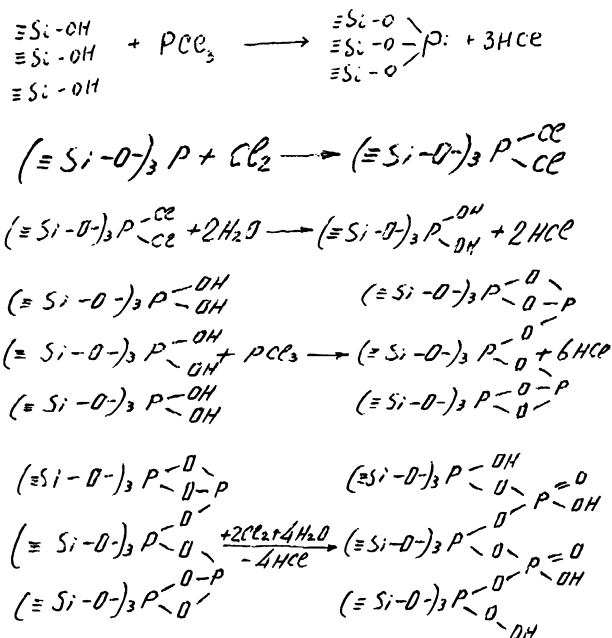


Fig. 13. The formation a phosphorus (V) oxide layer on silica gel by the ML method using redox reactions. Reproduced with permission from candidate dissertation of A. N. Volkova. Copyright 1969, LTI by Lensovet.

The study of catalytic, sorption, electro-physical properties of films obtained by ML with different chemical composition and structure revealed a number of important fundamental observations related to the number of ML cycles.^[15,16,33,59,69,79,80] These observations include: the “monolayer effect” which observed sharp change in the material properties of the surface after just 1-4 ML cycles had been applied; the substrate shielding effect that observed complete physical coverage of a matrix surface after at least 4-6 ML cycles; effects related to multi-component systems; and effects related to the mutual structural coordination between a substrate surface and an ML thin film.

“The monolayer effect” was first mentioned in V. M. Smirnov’s 1973 candidate thesis on the study of catalytic properties of ML-synthesized metal oxide monolayers on silica gel surfaces.^[55] He observed a two orders of magnitude increase in catalytic activity K_s (specific rate constant) of silica gel after it was coated with just a monolayer of titanium oxide. This increased catalytic

activity was for a model gas-phase reaction of carbon tetrachloride hydrolysis. Upon further ML film deposition to 4 then 6+ layer, K_s gradually decreased to values typically observed for bulk titanium oxide with anatase structure (cf. shielding effect). Monolayer and the multi-component system effects were observed in multi-layered phosphorus oxide and titanium oxide ML films on silica gels. A variety of alternating layers were tested including: P, PT, PTPT, T, TP, TPT and others, where P is a monolayer of phosphorous oxide and T is a monolayer of titanium oxide.^[56] Samples where phosphorus oxide monolayers were the final cycle at therefore at the surface (sample P, TP, PTP, TPTP) completely lost activity for carbon tetrachloride hydrolysis. Meanwhile, where titanium oxide monolayers were deposited last, and were on the surface (samples T, PT, TPT, PTPT), samples showed increased activity.

Before concluding the brief history of research before the end of 1974, it should be noted that this period included many important experimental steps after the original “framework” hypothesis. Also it should be noted that in Soviet Union, based on work from LTI by Lensovet at the Department of Chemistry of Solids, a new field of ML research was created which confirmed the broad synthetic capabilities and applications of the ML method. The main scientific works on the ML method from this period (pre-1974) have been published in Soviet scientific journals (many translated to English) and in novel materials patents.^[70-90]

It is also important to explain why 1974 is the cut-off date for discussion of the ML method in this section of the paper. With priority 1974, a Finnish patent by T. Suntola was published in 1977. It described alternate deposition of thin monolayers under high vacuum conditions. This “new” method by Suntola became known as atomic layer epitaxy (ALE).^[14] In the recently published article on the history of ALE and ALD the author, R. Puurunen, wrote that research on atomic layer deposition (ALD) began with Suntola’s patent.^[21]

5. 1975 to 1980s: The Beginning of Active ML/ALD Research and Development Across the World

The 15-year period from 1975 to the late 1980s was when molecular layering was decisively adopted and recognized

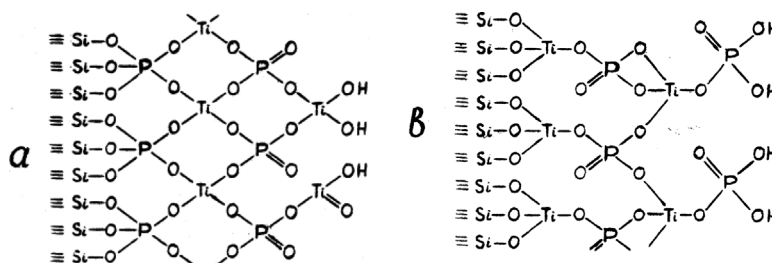


Fig. 14. The chemical structure of the products of molecular layering of phosphorus and titanium oxides on silica gel: a) PTPT sample; b) TPTP sample. Reproduced with permission from candidate dissertation of A. N. Volkova. Copyright 1969, LTI by Lensovet.

as a new method of materials synthesis. This recognition came from scientists of different countries and saw the emergence of a significant number of publications not only in the USSR, but also in Bulgaria and the German Democratic Republic (GDR).^[23–28]

An increased number and variety of solid-phase matrices (substrates) were studied with the ML process during this period. Researchers actively began to use metal organic precursors with halides. This significantly expanded the range of materials that could be synthesized with ML. Many studies at that time were application-oriented, and new ML equipment and instrumentation was developed. This period produced the first ML materials at scale for industrial applications.^[42,91–140] This time period can be divided into two stages: before and after 1979. In 1979 T. Suntola and his co-authors filed a patent (issued 1983) that related to zinc sulfide film deposition by ALE.^[31] This patent, unlike the 1977 patent, shows zinc sulfide film deposition on glass substrates alternating doses of zinc chloride and hydrogen sulfide precursors. Removal of excess reagents and gaseous reaction byproducts occurred after each dosing. This patent used molecular precursor as opposed to his earlier with elemental precursors. Soon after, an article by A. Aildla and colleagues^[141] described the claimed principles of the ALE method. These claimed principles are in essence the same as the ML principles set out in S. I. Kol'tsov's doctoral dissertation and his pre-1979 publications.^[32,33,43] It is important to note that this article was perhaps the first information for the Russian scientists about the ALE method.

As his patents became widely known, T. Suntola presented his work at numerous international conferences. He received a lot of publicity and the non-Russian scientific community began to view him as the creator of this new thin film method. Certainly, T. Suntola and his colleagues made significant contributions to the development of the ML/ALD method including the use of metal organic precursors, the subsequent introduction of the technique to industry, and the development of process equipment. It is his work that we believe gave a new motivation to the practical applications of the ML/ALD method and also attracted the attention of many other developed countries to the new technology.

At the same time, parallel to Suntola's work, and up until the second Finnish patent of 1979^[31] a number of V. B. Aleskovskii and S. I. Kol'tsov's students also contributed significantly to the ML field. Candidate degrees were received by: V. B. Kopylov (1977), A. A. Malkov (1977), M. A. Ereemeeva (1977), V. I. Kovalkov (1977), N. A. Stepanova (1978), A. M. Postnova (1978), V. E. Drozd (1978), V. K. Gromov (1979), V. N. Postnov (1979) (Table 1). Most of their work was published in journals.^[90–93,95–97,102,121] ML-related publications by Bulgarian and German scientists on the subject of catalysis should also be mentioned here.^[23–28,142–144] In the following years, from 1980 to 1990, about thirty candidate degree theses were defended in the ML field in the Soviet Union. The most important of these works are

shown in Table 1. Most of the results were published in journals.^[65,98,99,103,106–111,113,120,124–125,127–129,132–138,145]

As can be seen from the titles of the theses, research areas in the 1980s were not only fundamental but also highly applied. To study ML films, M. A. Ereemeeva first used IR-ATR spectroscopy, S. E. Kurashvili used quartz crystal resonators for “in situ” sample mass change monitor during deposition process, and A. I. Kudryashova used techniques to evaluate the type and strength distribution of acid-base centers on a surface.^[90,127,135] Most of the works from this period deal with the functional properties of the resulting ML materials (V. B. Kopylov, A. A. Malkov, V. I. Kovalkov, A. M. Postnova, V. E. Drozd, V. N. Postnov, M. N. Tsvetkova, A. V. Brykalov, A. A. Seytmagzimov, E. Y. Popova, V. D. Kupriyanov, E. A. Avrutina, S. A. Trifonov and others)^[91–110, 115, 120–134, 136–138]. The following can be considered pioneering works in the field: A. V. Evdokimov's development of synthesis techniques based on ML principles where synthesized monolayers contained heterogeneous atoms,^[128] S. A. Trifonov's studies on the first modification of polymeric materials with the ML method,^[110] T. A. Vitkovskaya's modification of lumino-phores with ML,^[120] E. A. Avrutina and E. Y. Popova's use of the ‘ML film overcoat effect’ to create shell pigments and composite material fillers,^[106,136] V. E. Drozd's first vacuum ML equipment and subsequent semiconductor and dielectric coatings, as well as a number of other works.^[113,114,131]

During this period (1973–1990) chemical assembly of carbon structures on solid surfaces was further developed, as was various carbon material surface modifications, initiated by E. P. Smirnov.^[88,89,99,108,109,116,146,147] For example, research was conducted on ML synthesis of carbon coatings on silica gel, on diamond, on carbon fiber (see candidate dissertations of V. B. Kopylov (1977), S. K. Gordeev (1980), V. D. Ivin (1980) (Table 1). This was achieved by alternating surface reactions between methane and carbon tetrachloride on the surface of specially pretreated for example chlorinated diamond powder. A linear increase in carbon content was observed with number of ML cycles indicating uniformity of the carbon build-up process.^[99] Fig. 15 shows the process of formation of carbon structures on the surface of carbon fiber.

Discussions by V. B. Aleskovskii about the ML method's capabilities at the Academy of Sciences of the USSR and S. I. Kol'tsov's experimental results convinced V. B. Aleskovskii focus applied research on microelectronics. This idea was actively developed by M. V. Keldysh who was president of Academy of Sciences of the USSR at that time. One can only wonder about wisdom of these individuals who were many years ahead of actual ML/ALD development for microelectronics.

One of the important tasks at the time was to obtain metal-insulator-semiconductor (MIS) structures with dielectrics deposited by ML. The dominant technology at that time was complementary metal-oxide-semiconductor (CMOS) technology. However, at that time, the thicknesses

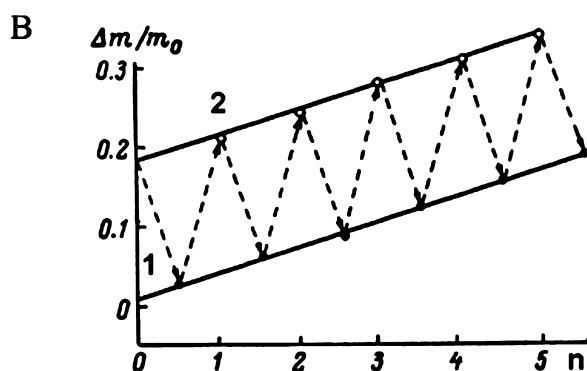
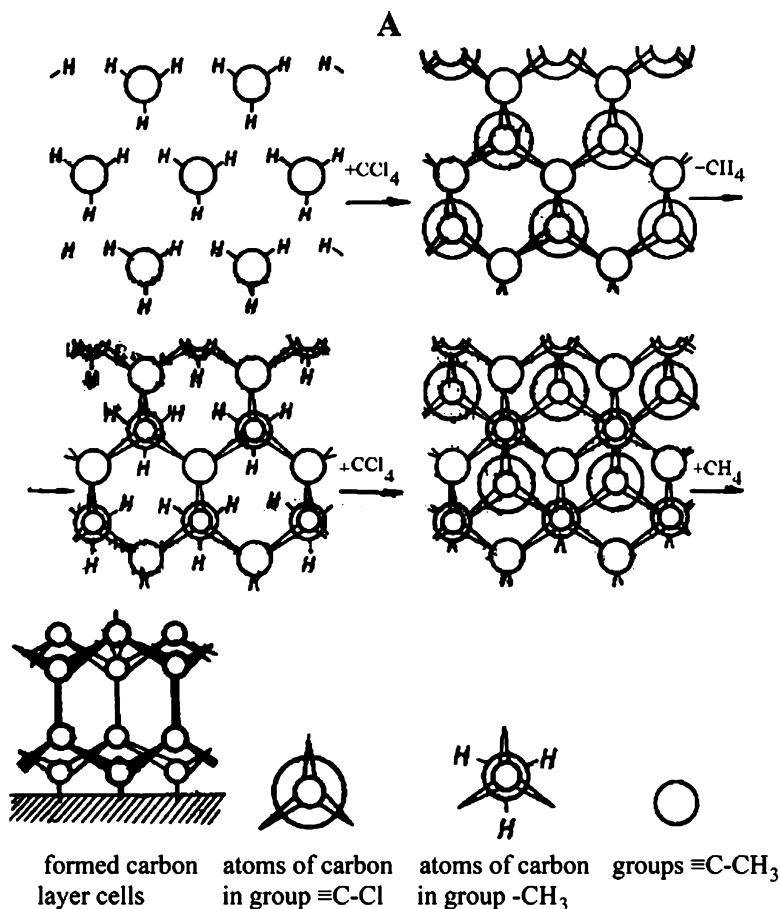


Fig. 15. Synthesis of carbon layer by atomic layer deposition on the surface of carbon fiber: A - reaction scheme; B - mass change as a function of cycle number ML (n) in step chemisorption of methane (1) and carbon tetrachloride (2). Reproduced with permission from ref. [184] and from candidate dissertation of V.D. Ivin. Copyright 1980, LTI by Lensovet.

of dielectric gate oxides were tens of nanometers, and the necessity for higher- k films was not yet present in the Soviet Union or abroad. However, there was an interest in thin dielectrics film for MIS structures from a fundamental scientific point of view as well as for a couple of niche applications. In the mid-1970s, in cooperation with the one of the leading USSR electronics engineering organizations, “Svetlana”, the possibility of ML-made dielectric thin films for metal-oxide-semiconductor (MOS) structures was demonstrated for the first time. These high- k materials

included chromium, titanium, vanadium, hafnium, zirconium, tantalum, niobium and molybdenum oxides.^[100]

In the following years, studies were conducted to obtain films without the use of water as a precursor. An example is multilayer periodic structures based on II-VI compounds, such as ZnS, CdS, ZnSe, CdSe and CdTe.^[134,137,148–152] It was important to show the capabilities of the ML method in the early 1980s to grow ultrathin superlattice structures. For these studies, quartz crystal microbalance (QCM) was installed into the reactor to study the reaction mechanisms

in situ.^[150–152] See the dissertation of S. E. Kurashvili (1984, Table 1) for more details.

Previous work on basic and applied thin film research allowed ML-based production of new industry-applicable materials in the mid-1980s. In the USSR, these were the first industrial products made by the ML method. These products were silica gel modified with phosphorus oxide layers^[82] (sorbent FS-1-3 for water vapor, amines and organic compounds) and vanadium oxide monolayer^[94,119] (sorbent-indicator IVS-1 for water vapor). There were also new sorbents that were industrially available and used to stabilize the gas media in the hermetically sealed aviation equipment and other industrial areas.^[104] One of the devices which employs IVS-1 and FS-1-3 materials is shown in Fig. 16.

The introduction of these and other new materials made by ML gave motivation to the development of both laboratory and industrial pilot-scale installations.^[117,118,126,139,140]

Studies by V. B. Aleskovskii's school on the technological equipment design for the chemical assembly of materials by ML can be divided into three main research areas: 1) flow technology for deposition under atmospheric pressure, mostly for modifying highly porous, dispersed materials (inorganic and organic fillers, pigments, porous sorbents, substrates for heterogeneous catalysis, powders for synthesis of ceramics, fibers, etc.), with high surface areas and high surface to volume ratios; 2) vacuum technology for semiconductor materials, thin films in nano-, micro- and optoelectronics, etc., i.e., processes which use objects as templates with large volume to the surface area ratios; and 3) combined flow-vacuum technology.



Fig. 16. Aviation flight director with sorbents IVS-1 and FS-1-3 installed. Photograph from Malygin's collections; published with permission of Anatoly Malygin (private collection).

Based on ML principles and related experimental knowledge of ML thin films (oxides, nitrides,^[64] carbon, sulfides, metals^[112]) on various surfaces (particles, fibers, plates, dielectric product, semiconductors, metals, glass, polymers, etc.), it is necessary in all cases to implement a certain sequence physico-chemical steps: a) Matrix surface preparation must be carried out to form the desired composition and location of functional groups. This can be done by heat treatment or other physical and chemical treatments including solution or gas phase chemical treatment, plasma, UV irradiation, laser radiation; b) Chemical reactions must proceed using appropriate, low molecular weight (high vapor pressure) precursors that react well with functional groups the surface; and c) Excess unreacted reagents and gaseous reaction products must be removed before dosing of the next gas-phase reactant.

To form a film with a given thickness the steps above must be repeated one after the other a desired number of times. The chemical composition, structure, the properties of the film is determined by compliance with the aforementioned steps. Chemical process parameters including temperature, reagent concentration, and the rate of the supply and removal of gaseous products will also affect the final film. When dispersed materials used as coating substrates, a significant impact on the deposition rate has a hydrodynamic regime in the system “gas–solid”. For example, the synthesis can be carried out in conditions of fixed, moving or fluidized-bed solids.

Among the works related to the design of ML equipment, research on vacuum ML equipment by V. E. Drozd should be noted. In the 1970s he was constantly improving the design of vacuum ML units. In 1975, he was the first person to make an ML system with “in situ” film thickness monitor system during deposition of films on polished silicon. This apparatus combined an ellipsometer and the Peltier element (Fig. 17).^[78] After release of the article of reference 131, V. E. Drozd developed and produced an automated

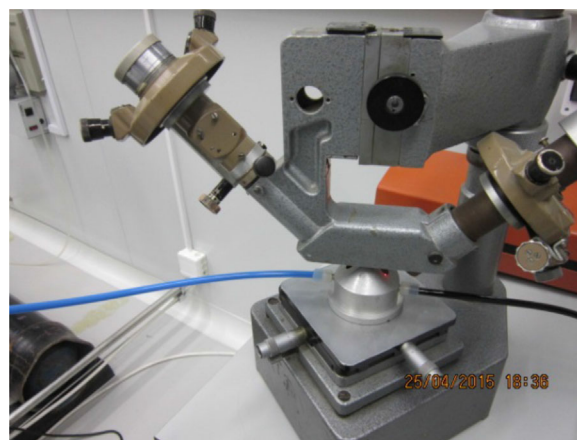


Fig. 17. Vacuum ML reactor combined with an ellipsometer (1975). Photograph from Drozd's collections; published with permission of Victor Drozd (private collection).

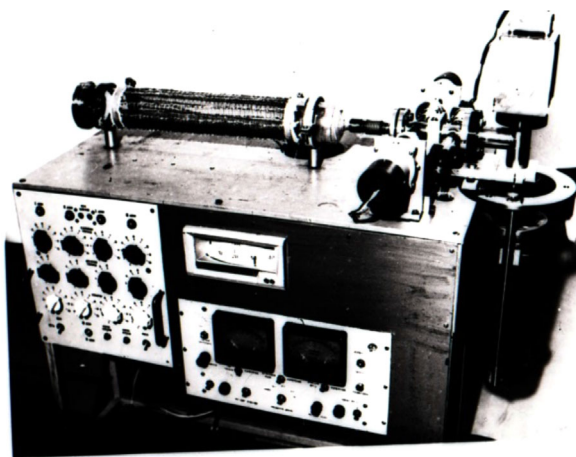


Fig. 18. Automatic ML installation with analog control system (1977). Photograph from Drozd's collections; published with permission of Victor Drozd (private collection).

ML setup with an analog control system (Fig. 18). In these automated vacuum-flow reactors, many metal oxides were synthesized including those deposited with low vapor pressure metal chlorides placed in specially heated cells connected to the reactor.^[96,100,101,113,153–156]

In the 1980s new flow type ML installations were created which served as prototypes for industrial scale apparatuses. V. P. Tolstoy and S. N. Gruzinov invented and tested a reactor for ML on large-sized objects (Fig. 19).^[126,139] ML layers are applied (Fig. 19) using a nozzle 1, which moves along the surface of the substrate at a given speed and at a certain distance from it. The substrate is not placed in a closed reactor, so its size is not limited. The patent shows an example of a

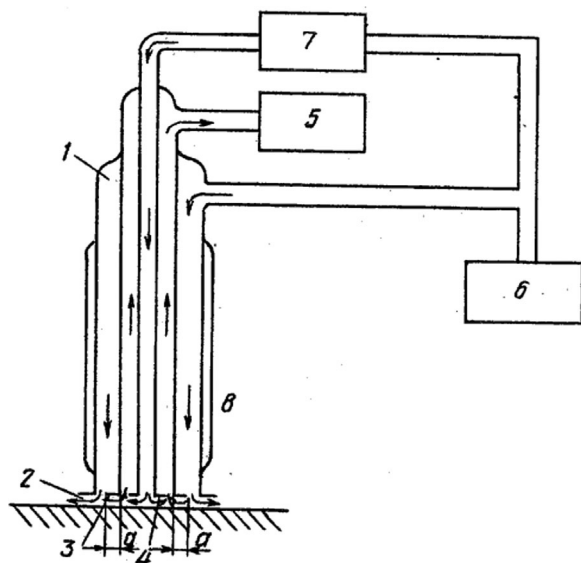


Fig. 19. Figure from Tolstoy's "tube in tube" ML reactor patents. 1) nozzle; 2) pipe for supplying dry air; 3) pipe for pumping reagents and dry air; 4) pipe for feeding reactants to the substrate surface; 5) pump; 6) feed system for dry air; 7) system reagent supply; 8) heater. Scheme from Tolstoy's collections; published with permission of Valery Tolstoy (private collection).

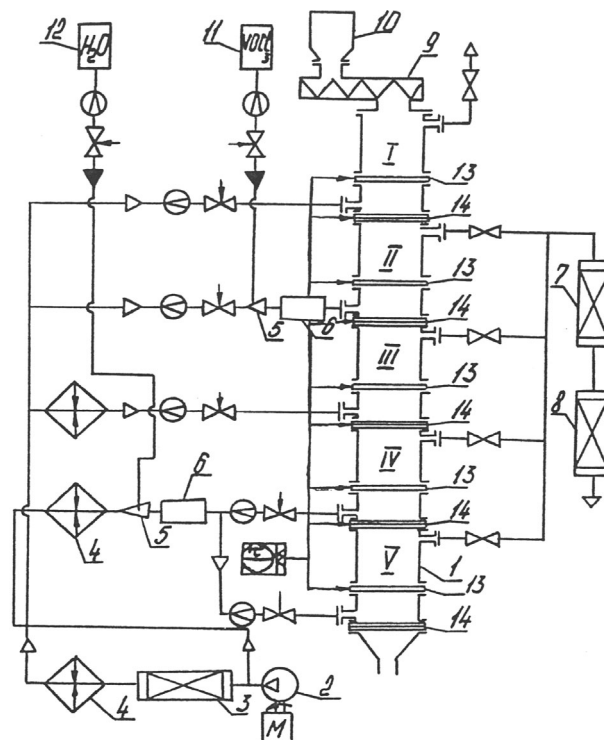


Fig. 20. Schematic representation of the apparatus for IVS-1 production.^[140] 1) ML reactor; 2) blower; 3) dehumidifier; 4) heater; 5) spray gun; 6) mixer; 7, 8) absorbers; 9) feeder; 10,11,12) tanks with silica gel, vanadium oxochloride and water; 13) gas distribution device; 14) isolating assembly. Reproduced with permission.^[140] Copyright 1987, LTI by Lensovet.

2500×1000×10 mm glass plate as the substrate. By regulating the feed rate of dry air (2-6) and reagents (4-7) a separation of reactants and reaction products is created, limiting each to a given area. Moreover, unwanted gases are removed from the reaction zone (3 through 5). Nodes 5, 6, 7 are connected to the nozzle through a flexible hose.

V. F. Dergachev and colleagues developed, manufactured and tested an automated multi-sectional ML apparatus for powder coating using a fluidized bed mode.^[117,118] Fig. 20 shows a diagram of the apparatus used for the ML synthesis of a vanadium-oxide monolayer on a silica gel surface ($\equiv\text{Si}-\text{O}-$)₃VO. As indicated above, this product was used in industry under the name IVS-1 as a high-sensitivity gas phase humidity indicator. As per Fig. 20, solids are discreetly transferred from the top column to the lower level columns where the silica gel gets sequentially treated by continuous upwards flow of: 1) dry air (for surface preparation); 2) VOCl₃ for chemisorption; and 3) dry and moist air for desorption/removal of unwanted reactants and side products. The semicontinuous equipment proposed in this work involves separation of the gas flow in the individual synthesis stages. Fig. 21 and 22 show installation photographs during the first experiments conducted by V. F. Dergachev under A. A. Malygin's supervision (in the picture).

Often there are questions about the relationship between molecular layering research groups in the USSR and other

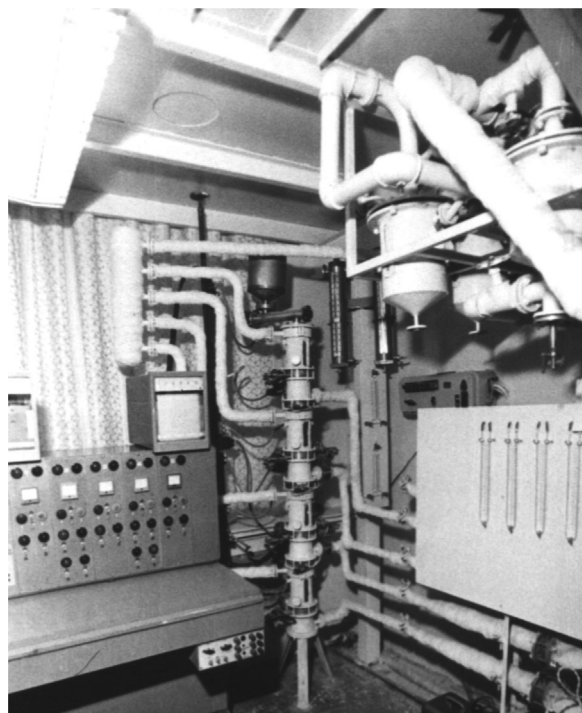


Fig. 21. Photography of pilot scale ML installation for production of IVS-1 (1982). Photograph from Malygin's collections; published with permission of Anatoly Malygin (private collection).

countries such as: Were there any other countries beside USSR where ML research was conducted? And how did the relationship between Soviet and foreign scientists in this field evolve? Because of the Soviet period, the first international contact between Russian ML researchers and others was with representatives of the Eastern European socialist countries. At that time, V. B. Aleskovskii, S. I. Kol'tsov, and their students met with colleagues from Bulgaria and East Germany who were engaged in ML studies and whose publications indicated familiarity with the Russian ML work.^[23–28,142–144] The Bulgarian scientist

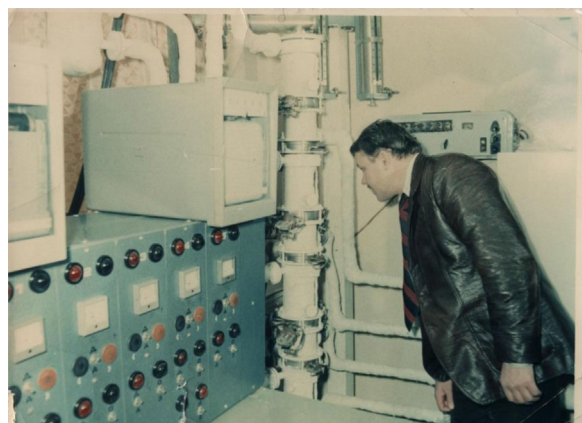


Fig. 22. The first experiments on the column flow unit (1982) (right – A. A. Malygin). Photograph from Malygin's collections; published with permission of Anatoly Malygin (private collection).



Fig. 23. Conference in Linz (Austria): first row left to right – T. Suntola, V. Drozd, L. Niinisto, M. Leskela; the person on the left turning to the camera is Professor S. M. George. Photograph from Drozd's collections; published with permission of Victor Drozd (private collection).

D. Damyanov from Burgas Institute of Technology visited the Department of Chemistry of Solids at the LTI by Lensovet in the 1970s.

By the early 1990s, the possibility for Russian scientists to participate at international conferences on topics related to ALE opened up. Fig. 23 shows a photograph taken at one such conference, where V. E. Drozd met with Finnish and other ML/ALE scientists. Thanks to information provided by Professor M. Pessa with the support of L. Niinisto, V. E. Drozd's first report in collaboration with V. B. Aleskovskii on molecular layering research in the USSR was presented at the ALE symposium (1st International Symposium on Atomic Layer Epitaxy, Espoo/Helsinki, Finland, June 13–15, 1990) in 1990.^[157] V. E. Drozd met with T. Suntola in Helsinki on the first symposium on the ALE and handed him an invitation from V. B. Aleskovskii to visit LSU. The same year, Leningrad State University invited T. Suntola to give a talk at the Solid State Chemistry Department where, one might say, an historic meeting with V. B. Aleskovskii took place. It should be noted that V. E. Drozd made significant contributions to the preparation and organization of this event. Doctor T. Suntola presented the work of Finnish researchers in the field of ALE including his own work on electroluminescent doped ALE zinc sulfide coatings.

After Suntola's presentation, many questions were asked. Most people were concerned about the instrumentation and the solution to environmental problems of ALE. Not many questions about the deposition methods and its features were asked. At the end the questions, T. Suntola said that he was little surprised that nobody asked about the fundamental aspects of the ALE method. To this, V.B. Aleskovskii answered that he and his team had developed the ML method more than 20 years ago, which Suntola called ALE in his talk. For many Soviet - Russian scientists involved in the study of ML that built the ML field from the ground up, it is still not completely clear how Finland, as a neighboring

country participating in related research for more than twenty years did not know anything about the work of V. B. Aleskovskii, S. I. Kol'tsov, and their students.

In the 1980s, fundamental ML work focused on better understanding materials produced with ML. During this period, and for the first time since S. I. Kol'tsov's 1971 doctoral dissertation, the first doctoral thesis by V. N. Pak (1982, Table 1) was published. This thesis presented detailed studies on the structure of oxide coatings synthesized by ML employing quantum-chemistry approaches. A little later there was a related doctoral dissertation in Bulgaria, defended by D. Damyanov (1988, Table 1). This dissertation was devoted to the preparation of catalysts by ML.

Thus, research between 1975 and 1990 demonstrated both the numerous synthetic capabilities of ML and also how ML can make materials with various functional purposes. This period saw creation of the technological basis of the ML method, its hardware, and the first ML synthesized materials used in industry. During these years, the geography of the ML/ALD research was significantly expanded thanks to the work of the Finnish scientists under the guidance of T. Suntola. Suntola's work implemented industrial production of electroluminescent displays enabled by the new technology. Since 1990 the research of ML/ALD has developed rapidly in many countries and work on ML/ALD is regularly presented at international conferences by a variety of international scientists. We can say that the period between 1975 and 1990 was when the international scientific community of researchers started actively and frequently share information about the new developments in the field of ML/ALD. In this same period research in the field of ML/ALD started to receive international recognition.

6. From 1991 to 2015: The Latest Scientific Achievements and the Creation of the International Community in the Field of ML/ALD

Since 1990, research on ML/ALD in Russia has been propelled by rapid developments in nanotechnology and nanomaterials. Indeed, in accordance with modern scientific concepts, ML/ALD is a typical "bottom - up" nanotechnological process able to coat practically any solid (including nanoscale solids) with defined, chemically assembled nanostructures or films.

A characteristic feature of the past 25 years of development in the field of ML in Russia is that V. B. Aleskovskii's and S. I. Kol'tsov's students formed their own research areas reflected by their doctoral dissertations. These subsequent works include the following doctoral dissertations: A. A. Malygin (1991), G. U. Ostrovidova (1992), V. M. Smirnov (1994), A. P. Nechiporenko (1995), V. V. Gusarov (1996), A. G. Syrkov (1998), Y. K. Ezhovskii (2002), V. B. Kopylov (2007), V. P. Tolstoy

(2009), M. M. Sychev (2013). We would like to particularly note the doctoral thesis by a graduate of the Department of Chemistry of Solids at LTI by Lensovet, A. P. Alekhin (1999), who in the early 21st century was the director of the leading electronics intuitions of USSR - Research Institute of Physical Problems by N. F. Lukin, in the city of Zelenograd (Moscow region) and then heading the Department of General Chemistry in the prominent Moscow Institute of Physics and Technology (state university) (MIPT).³ ALD research at MIPT is currently headed by candidate of sciences A. Markeev. A significant contribution to the research on carbon composite materials was made by another graduate of the solid state chemistry department, Dr. S. K. Gordeev who heads a Laboratory of the Joint-Stock Company "Central Research Institute of Materials" (Saint Petersburg, Russia). In St. Petersburg State Pedagogical University, under the supervision of one V. B. Aleskovskii's Ph.D. students V.N. Pak, and in collaboration with Dr. V. B. Kopylov, G. U. Ostrovidova has been conducting research in the field of chemical transformations of solids and molecular layering. In recent years a number of new Russian scientific groups involving ML/ALD research have been formed in the following institutions: St. Petersburg State Electrotechnical University "LETI" by V. I. Ulyanov (Lenin) (led by professor V. V. Luchinin), Stavropol State Federal University, St. Petersburg State Polytechnic University, Ioffe Physical-Technical Institute of the Russian Academy of Sciences, Mordovian State University (city of Saransk), and an ALD group in Dagestan State University (Makhachkala) created with the assistance of A. I. Abdulagatov. There are publications from a number of scientists at universities and institutions from Tomsk, Novosibirsk and Kazan. Russian vacuum equipment manufacturing company "Ferry Watt, LLC" developed a new ML/ALD thin film apparatus for coating both semiconductor wafers and also nanoparticles.^[158]

The work performed between 1978 and 2015 has been a significant contribution to the emergence of new ML/ALD research centers in Russia. This has been aided by the recent dissemination of information and experimental results relating to V.B. Aleskovskii's scientific school "Chemistry of highly organized substances". Promotion of the ML method and its accomplishments and prospects for both fundamental research and practical application has also come from the many ML/ALD works of non-Russian scientists. The well-known virtual project on the History of ML/ALD contributed to the creation of the international ML/ALD community with participation from Russian scientists evidenced by addition of their abstracts submitted by R. Puurunen in 2014 to the ALD conferences in Helsinki and Kyoto.^[19,20] More than 30 scientists from ten countries are co-authored on these publications. Further strengthening of the relations between scientists from

³ MIPT (department of A. P. Alekhin) was the first in Russia who in 2007 bought the foreign ALD device

different countries contributed the International workshop “Atomic Layer Deposition: Russia, 2015”, held on 21–23 September 2015 (Moscow, Dolgoprudny, MIPT) and the 13th International Baltic Conference on Atomic Layer Deposition – Baltic ALD 2015, 28–29 September 2015 (Tartu, Estonia).

Despite the emergence of new scientific groups studying ML/ALD, the main centers of ML in Russia are the original groups created by V. B. Aleskovskii in St. Petersburg State Technological Institute (Technical University) and St. Petersburg State University. These universities have accumulated five decades of research experience in the field of solid synthesis by ML. Therefore it is not surprising that doctoral dissertations from these universities show new directions for both basic and applied research into solid state material science. The Department of Chemical Nanotechnology and Electronic Materials (CNME) under the direction of Prof. A. A. Malygin, continues both fundamental and applied ML research in areas of solid phase material science including: inorganic sorbents, catalysts, ceramics, polymer composites, sensors, etc.^[80,81] Under Malygin's supervision, a number of automated ML apparatuses have been developed for: depositions of functional thin films on quartz for fiber-optic sensors applications (Fig. 24); glass bodies for X-ray tubes (Fig. 25); and new apparatus for the production of ML-based supercapacitors (Fig. 26).^[159–162]

Many works focused on the commercialization of ML technology in cooperation with industrial partners. For example, materials nanodoped by ML and used in producing ceramic insulators for X-ray tubes (Fig. 27) have been developed in collaboration with “Svetlana-Roentgen” company. Development and production of ML equipment for previously mentioned sorbents IVS-1, FS-1-3 and modified ceramic blends for composite materials used by industry, is carried out in “Chemical Assembly of Nanomaterials, LLC” (Saint Petersburg, Russia).



Fig. 24. A flow-vacuum automated system for deposition of functional thin films on quartz fibers used for manufacturing of fiber-optic sensors (2011). Photograph from Malygin's collections; published with permission of Anatoly Malygin (private collection).



Fig. 25. A flow-vacuum, automated ML installation based on an industrial vacuum oven for deposition of functional nanocoatings on glass tubes for X-ray applications (2014). Photograph from Malygin's collections; published with permission of Anatoly Malygin (private collection).

The team of the Chemical Nanotechnology and Electronic Engineering Materials Department at SPbSTI consists of its graduates and candidates of sciences: A. A. Malkov, N. V. Zakharova, V. F. Dergachev, E. A. Sosnov, S. D. Dubrovenskii (Table 1); young scientists and graduate students: V. V. Antipov, S. V. Mikhailovsky, N. Yu. Efimov,



Fig. 26. Project of a new flow-vacuum ML unit “Ales-1” for the manufacture of supercapacitor components (joint venture between SPbSTI and “Chemical Assembly of Nanomaterials”, LLC) (2014). Photograph from Malygin's collections; published with permission of Anatoly Malygin (private collection).



Fig. 27. Nanodoped blend for ceramic insulators used in X-ray tubes (2010). Photograph from Malkov's collections; published with permission of Anatoly Malkov (private collection).

I. S. Bodalev, E. O. Drozdov, A. S. Kochetkova, who in the next few years will defend their candidate dissertations. A number of promising young candidate of sciences (graduates of the same department): O. V. Osipenkova (1998), V. A. Lapikov (2002), S. A. Belova (2006), A. V. Kutchiev (2007), A. K. Bulkina. (2009), K. S. Anisimov (2012), Yu. M. Koshtyal (2012), A. N. Gukova (2013) successfully work in other organizations including foreign companies.

Important areas of research related to the studying temperature effect on ML when ceramic oxide or oxygen-free materials are used as a matrix (substrate) are led by A. A. Malkov, the main developer of the nanodoped blend for ceramic insulators.^[163] Fundamental research on the surface processes occurring during ML using quantum chemistry modeling tools was initiated and is currently led by S. D. Dubrovenskii.^[164–166] There are two candidate degree theses that have been defended on this subject (A. V. Kutchiev, A. N. Gukova) and there is a doctoral thesis in preparation to be defended in 2015–2016. Our applied research in the field of polymeric materials, sensors, alternative energy sources has significantly expanded in recent years.

Professor Yu. K. Ezhovskii works of the synthesis of “metal - dielectric - metal” and “metal - insulator - semiconductor” heterostructures and has identified conditions needed for constant oxide and nitride film growth rate (during nucleation) on the surface of single crystal silicon and gallium arsenide. He also demonstrated the applicability of the “free energy linearity principle” in his studies of the reactivity of solids.^[64,107,125,132,137] Since the 1980s research, led by S. A. Trifonov (including works of A. K. Bulkina, V. A. Lapikov, N. Yu. Efimov) has been on the regulation of polymeric and composite material properties by ML.^[167–169] Under the leadership of candidate of sciences E. A. Sosnov, original methods based on scanning probe microscopy (SPM), have been developed to

study the surface of dispersed solids, fibers, and bulk materials modified by ML.^[170–173] In 2007, the CNEEM department celebrated its 40th anniversary and published a collection of papers, which reflected the key achievements of their research teams in the development of the ML method.^[174]

Important and promising areas of molecular layering are still under development at the Department of Solid State Chemistry of the St. Petersburg State University supervised by V. B. Aleskovskii's and S. I. Kol'tsov's students. Professor V. M. Smirnov and his students have begun to study the connection between magnetic properties of solids and their atomic structures and composition. They have produced a number of original methods for this work. Since 2000 the Institute of Chemistry of St. Petersburg State University, under the guidance of Prof. V. M. Smirnov, has carried out studies on nanostructuring during synthesis of composite nanomaterials. Basic theoretical and experimental aspects of the chemical nanostructuring are considered in reference.^[175] These results are an important contribution to the development of scientific foundations of a new generation of composite materials (ceramic, metal, polymer matrix) with a given structure and adjustable functional (mechanical) properties.^[176]

Significant contributions to the development of vacuum ML equipment, in particular for nanoelectronics, have been, and continue to be, made by a group of students under the guidance of the candidate of Physical and Mathematical Sciences, V. E. Drozd. He first obtained the superlattice of thin chalcogenide films by ML. As V. E. Drozd noted number of publications on the synthesis of CdTe films by ML is very limited and he was interested in developing a fast-track technology for CdTe synthesis. He developed a special thin film apparatus (Fig. 28) that allowed deposition of high-quality CdTe film with, unusually for ML/ALD, a high growth rate of up to 5 microns per hour.^[177–179]



Fig. 28. ML apparatus for accelerated growth of CdTe (1998). Photograph from Drozd's collections; published with permission of Victor Drozd (private collection).

Developed under his direction, ML equipment has since been purchased by a number of universities and research centers. According to V. B. Aleskovskii, Professor V. P. Tolstoy created spectroscopy of nanostructures, developed an original method of ion layering, and offered one of the first ML installations for thin film coatings of large products.^[126,139,180] Candidate of sciences V. N. Postnov obtained highly effective catalysts and sorbents by ML and recently used template synthesis techniques in combination with the ML method to create complex organic systems on the surfaces of inorganic and carbon carriers.^[181]

In 2003, the Solid State Chemistry Department at St. Petersburg State University celebrated its 25th anniversary (see. Fig. 5). A collection of scientific research articles was published by the department faculty, which reflected all the main areas of department research.^[181] In this department proceeding, V. B. Aleskovskii marked the names of many of the participants of works in the field of solid state chemistry and molecular layering: A. I. Romanychev, I. Gimanova, Yu. M. Artemyev, L. P. Bogdanov, A. G. Novikov. The list could be continued to including E. G. Zemtsova, O. M. Osmolovskaya and many others. Students in the departments of SPbSTI and SPbSU study different specialties, but they definitely listen to lectures, perform laboratory classes in chemistry of solids, and study the method of molecular layering. V. B. Aleskovskii, S. I. Koltsov and their colleagues published textbooks, manuals, methodical instructions to laboratory works, which students use in the educational process.^[34,35,140,182–184,185–187]

Today the works of researchers at SPbSTI and SPbSU have annual financial support, not only from various Russian scientific funding agencies (Russian Foundation for basic research, Russian Scientific Fund), but also from collaborations with industry in Russia and in other countries such as Germany, France. V. B. Aleskovskii's ideas would not always be understood by his early colleagues. In the 1960–1970s many scientists in the Soviet Union did not understand the essence of the “framework” hypothesis. His views on the possibility of obtaining individual solids with stoichiometric composition were not supported as many did not believe in the capability of molecular layering to deposit nanostructured films with atomic-molecular precision. At the 2004 conference in St. Petersburg State University, A. A. Malygin talked with Valentin Borisovich and asked him why (in the seventies) many of his ideas were not supported by the scientific community. V. B. Aleskovskii answered with some bitterness, but with a smile, that unfortunately his ideas often begin to be understood by others 20–30 years after they are originally conceived (see. Fig. 29). He had the right to say so because he was a productive member of the scientific community for more than 70 years, and despite the fact that he was an elderly man in his later years, he still actively continued his work generate new ideas and experiments. Time sorts things out and puts everything in its place.



Fig. 29. In the winter garden of the St. Petersburg State University during the break at the conference “Chemistry of highly organized substances and fundamental scientific basics of nanotechnology” (2004, V. B. Aleskovskii (left) meets his student Professor A. A. Malygin). Photograph from Malygin's collections; published with permission of Anatoly Malygin (private collection).

7. Conclusion

Concluding the brief history of the development of the molecular layering method in the Soviet Union and Russia, it can be noted that finally, at the beginning of the 21st century, an international ML/ALD community has been formed. Development of ML started in the middle of the last century with a small team of scientists at the LTI by Lensovet. Present day results show that this ML/ALD technology has made its way solidly into industry, and hundreds of laboratories, both in Russia and abroad, routinely perform research in this new, dynamically growing area of science and technology. Currently, there is no doubt that this field, proposed in 1952 by V. B. Aleskovskii's fundamental “framework” hypothesis has great prospects and practical value and that the achievements of Finnish colleagues under guidance of Dr. T. Suntola have also been a great contribution. Thus, a small stream of ML/ALD knowledge has been continuously pouring into a powerful and deep river of research in chemistry and technology of the surface science of solids.

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